# RAHUL SHARMA

**Software Engineer | Full Stack, Backend & AI Applications**

+91-9105989864 | rahulsharmajan20@gmail.com
linkedin.com/in/rahu-ul | github.com/rahu-ul | leetcode.com/u/rahulsharmaglb

## EDUCATION

**B.Tech, Computer Science (Artificial Intelligence & Data Science)** **2023 – 2027**
GL Bajaj Group of Institutions (AKTU) **CGPA: 7.1 / 10.0**

## SUMMARY

Software engineer with strong skills in Java, DSA, and full-stack development. Built and deployed MERN applications and a multithreaded Java key-value store. Experienced in developing AI-powered features using RAG and semantic search. Solved 600+ DSA problems and contributed to open-source projects.

## TECHNICAL SKILLS

**Languages:** Java, JavaScript (ES6+), Python, SQL, HTML5, CSS3
**Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT, RBAC, Bcrypt, WebSockets (Socket.IO), Multithreading, TCP Sockets
**Frontend:** React.js, Redux Toolkit, Tailwind CSS, Responsive Design, Flexbox, CSS Grid
**Databases:** MongoDB (Mongoose), MySQL, ChromaDB (vector DB), Caching (LRU, Cache-Aside)
**AI/GenAI:** Gemini API, RAG, Embeddings, Semantic Search, Content-Based Recommendation
**Cloud & Tools:** AWS (EC2, S3), MongoDB Atlas, Vercel, Render, Cloudinary, Git, GitHub, Postman
**CS Fundamentals:** Data Structures & Algorithms, OOP, Operating Systems, DBMS

## PROJECTS

**RapidKV – Redis-Inspired In-Memory Key-Value Store** *Java, TCP Sockets, Multithreading* Feb 2026 – Mar 2026 | GitHub

- **Concurrency:** Engineered a **multithreaded Java TCP server** supporting concurrent multi-client access.
- **Data engine:** Implemented **custom data types**, **TTL-based key expiry**, and **LRU cache eviction**.
- **Reliability:** Built **AOF persistence with crash recovery** and graceful shutdown handling.
- **Performance:** Benchmarked at **57,956 ops/sec** (50K operations, 25 concurrent clients) with **1.74 ms P99 latency**.

**Shopora – Full-Stack E-Commerce Platform with AI Features** *React, Redux Toolkit, Node.js, MongoDB, FastAPI, ChromaDB, Gemini* Dec 2025 – Present | Live | GitHub

- **Full-stack:** Built a MERN e-commerce platform with **JWT authentication**, **RBAC**, admin panel, Stripe checkout, and REST APIs.
- **Deployment & performance:** Implemented **Socket.IO order tracking** and deployed frontend on Vercel, backend on Render, and database on MongoDB Atlas.
- **AI/GenAI:** Developed **AI features**, including a **RAG shopping assistant**, semantic search, recommendations, review summarization, and product comparison using FastAPI, ChromaDB, and Gemini.

**PlayTube – Video Streaming Platform with Custom Caching** *React, Redux, Node.js, MongoDB, RapidKV, AWS EC2* Oct 2025 – Nov 2025 | Live | GitHub

- **Full-stack:** Built a **MERN** video streaming app with a **React + Redux** frontend, user authentication, and **REST APIs** tested with Postman.
- **Media & deployment:** Implemented **Cloudinary** media handling and **deployed the application on AWS EC2**.
- **Performance:** Integrated my own **RapidKV** as a **Cache-Aside layer over TCP**, reducing query latency from **244.71 ms to 23 ms**.
- **AI/GenAI:** Integrated the **Gemini API** to power **semantic search** across video metadata.

## PROBLEM SOLVING, OPEN SOURCE & ACHIEVEMENTS

- **600+ DSA problems** solved on LeetCode and other competitive platforms.
- **GSSoC 2025 (MannSakha Chatbot): 2 merged PRs** adding **dark mode with theme toggling** and **responsive layouts** (Flexbox, CSS Grid, media queries).
- **Smart India Hackathon (SIH) 2025:** Waitlist Rank 6 | **QubitX 2025 Hackathon:** Finalist among 500+ engineering teams.